Subject: bio303 (past papers)

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Q: 1 what is half reaction?

Ans.1:- Half-Reactions: A half reaction is either the oxidation or reduction reaction component of a redox reaction. • A half reaction is obtained by considering the change in oxidation states of individual substances involved in the redox reaction. Although oxidation and reduction occur together, we consider the two halves of an oxidationreduction reaction separately.

• For example, the oxidation of ferrous ion by cupric ion is written as follows, Fe +2 + Cu +2 ↔ Fe +3 + Cu+ • Same can be described as 2 half reactions in the following form: 1. Fe +2 ↔ Fe +3 + e- 2. Cu +2 + e- ↔ Cu +

Q 2: Define pH. According to the equations on 25 degree centigrade?

Ans.2:- The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the [H+], i. e, the negative logarithm of the [H+] • pH = log 1/[H+] • = -log[H+] • The pH of water at 25°C, in which H+ and OH– ions are present in equal numbers, is 7.0 • pH = -log[1 x 10-7]= 7 • The symbol p denotes "negative logarithm of" • For each pH unit less than 7.0, the [H+] is increased tenfold; • for each pH unit above 7.0, it is decreased tenfold

Q 3 : competitive inhibition?

Ans.3:- Competitive Inhibition : • This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e. active site and, therefore, competes with the substrate for that site. Reversible inhibitors bind to enzymes through non covalent bonds. • Dilution of the enzyme-inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity

. Q5: write the mechanism of enzyme action?

Ans.: Mechanism of Enzyme Action Enzymes bind and chemically transform other molecules— they catalyze reactions. • The molecules acted upon by enzymes are called reaction substrates rather than ligands. • Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 105 to 1017. • The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the active site. • The surface of the active site is lined with amino acid residues with side chains that complement and bind the substrate and catalyze its chemical transformation. • Often, the active site encloses a substrate, sequestering it completely from solution. • The enzyme substrate complex, whose existence is central to the action of enzymes. • Enzymecatalyzed reactions have three basic steps: • binding of substrate: E+S↔ES • conversion of bound substrate to bound product: ES ↔ EP • release of product : EP ↔ E+P

Q 6)write the citric acid cycle?

Ans.: Citric acid cycle-1 • Also called Tricarboxylic Acid Cycle (TCA) or Krebs cycle (after its discoverer, Hans Krebs). • Three names for the same thing. • A series of chemical reactions used by all aerobic organisms to generate energy through the oxidation of acetyl CoA derived from carbohydrates, fatty acids and proteins into carbon dioxide, water and chemical energy in the form of adenosine triphosphate. Conversion of pyruvate to activated acetate • Cellular respiration and intermediates for biosynthesis. • Anaplerotic reactions to regenerate the acceptor • Conversion of acetate to carbohydrate precursors in the glyoxylate cycle.

Q.8) what is activation energy?

Ans.: The difference between the energy levels of the ground state and the transition state is the activation energy, G‡. The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction. • Reaction rates can be increased by raising the temperature,

thereby increasing the number of molecules with sufficient energy to overcome the energy barrier. • Alternatively, the activation energy can be lowered by adding a catalyst. • Catalysts enhance reaction rates by lowering activation energies. • The role of enzymes is to accelerate the inter-conversion of S and P. • i.e enzymes lower the energy of activation, Δ G‡, of a reaction. The enzyme is not used up in the process, and the equilibrium point is unaffected. • The role of enzymes is to accelerate the inter-conversion of S and P. • i.e.enzymes lower the energy of activation, Δ G‡, of a reaction. The enzyme is not used up in the process, and the equilibrium point is unaffected.

Q.9)Define Biogenetics.?

Ans.1: Bioenergetics is the quantitative study of the energy transductions changes of one form of energy into another—that occur in living cells, and of the nature and function of the chemical processes underlying these transductions.

Q.10: Decine Ionization.?

Ans.2:Ionization is the process by which an atom or a molecule acquires a negative or positive charge by gaining or losing electrons to form ions, often in conjunction with other chemical changes.

Q11)Competitive and noncompetitive inhibition?

Ans: Competitive Inhibition:- • This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e. active site and, therefore, competes with the substrate for that site. Reversible inhibitors bind to enzymes through non covalent bonds. • Dilution of the enzyme-inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity. Non-competitive Inhibition:-

• Inhibitors bind enzymes at sites distinct from the substrate-binding site and • generally bear little or no structural resemblance to the substrate. Non-competitive inhibition cannot be overcome by increasing the concentration of substrate.

Q12)What are isozymes ? Ans: Two or more enzymes that catalyze the same reaction but are encoded by different genes are called isozyme.

Q13)what you know about snRna.?

Ans :Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells. Most of them are complexed with proteins to form ribonucleoproteins They are distributed in the nucleus, in the cytoplasm or in both. • They are significantly involved in rRNA and mRNA processing and gene regulations

. Q14)what is hyperchromicity

Ans.Concomitant with this denaturation of the DNA molecule is an increase in the optical absorbance of the purine and pyrimidine bases—a phenomenon referred to as hyperchromicity of denaturation

Q15)pyurate three metabolism pathway ? Ans:Pyruvate is the end product of the second phase of glycolysis. • For each glucose molecule, two ATP are consumed in the preparatory phase and

four ATP are produced in the payoff phase, • This gives a net yield of two ATP per molecule of glucose converted to pyruvate. • each phosphoryl group, represented here as P, has two negative charges .

Q16)diffence b/w DNA and RNA 5).

Ans: Stands ForDeoxyriboNucleicAcid.RiboNucleicAcid.

Definition A nucleic acid that contains the genetic instructions used in the development and functioning of all modern living organisms. DNA's genes are expressed, or manifested, through the proteins that its nucleotides produce with the help of RNA.

The information found in DNA determines which traits are to be created, activated, or deactivated, while the various forms of RNA do the work.

Function The blueprint of biological guidelines that a living organism must follow to exist and remain functional. Medium of long-term, stable storage and transmission of genetic information. Helps carry out DNA's blueprint guidelines. Transfers genetic code needed for the creation of proteins from the nucleus to the ribosome. Structure Double-stranded. It has two nucleotide strands which consist Single-stranded. Like DNA, RNA is composed

of its phosphate group, five-carbon sugar (the stable 2-deoxyribose), and four nitrogen-containing nucleobases: adenine, thymine, cytosine, and guanine.

of its phosphategroup, five-carbon sugar (the less stable ribose), and four nitrogen-containing nucleobases: adenine, uracil (not thymine), guanine, and cytosine. Base Pairing Adenine links to thymine (A-T) and cytosine links to guanine (C-G). Adenine links to uracil (A-U) and cytosine links to guanine (C-G). Location DNA is found in the nucleus of a cell and in mitochondria. Depending on the type of RNA, this molecule is found in a cell's nucleus, its cytoplasm, and its ribosome. Stability Deoxyribose sugar in DNA is less reactive because of C-H bonds. Stable in alkaline conditions. DNA has smaller grooves, which makes it harder for enzymes to "attack." Ribose sugar is more reactive because of C-OH (hydroxyl) bonds. Not stable in alkaline conditions. RNA has larger grooves, which makes it easier to be "attacked" by enzymes. Propagation DNA is self-replicating. RNA is synthesized from DNA when needed. Unique Features The helix geometry of DNA is of B-Form. DNA is protected in the nucleus, as it is tightly packed. DNA can be damaged by exposure to ultra-violet rays. The helix geometry of RNA is of A-Form. RNA strands are continually made, broken down and reused. RNA is more resistant to damage by Ultra-violet rays.

17). Write down the laws of thermodynamics ?

Ans :The first law is the principle of the conservation of energy: it states that

for any physical or chemical change, the total amount of energy in the universe remains constant; energy may change form or it may be transported from one region to another, but it cannot be created or destroyed.

• Second law The second law of thermodynamics, says that the universe always tends toward increasing disorder: in all natural processes, the entropy of the universe increases

18). Write down two stages of pentose sugar ?

Ans:There are two phases of the pentose phosphate pathway: 1. The oxidative phase 2. The non-oxidative phase.

19)Three major fates of animals and vascular plant ?

Ans :In animals and vascular plants, glucose has three major fates: 1.It may be stored (as a polysaccharide or as sucrose). 2. Oxidized to a three-carbon compound (pyruvate) via glycolysis to provide ATP and the metabolic intermediates. 3. Oxidized via the pentose phosphate (phosphogluconate) pathway to yield ribose 5-phosphate for nucleic acid synthesis and NADPH for reductive biosynthetic processes.

20)the acceptor arm of tRNA.?

Ans:Arms or loops of tRNA • All tRNA molecules contain 4 main arms or loops. • 1-Acceptor arm: This is made up of unpaired sequences of cytosinecytosine- adenine (CCA) at the 3´end The 3´OH group of adenine binds with the carboxylic group of a specific amino acid and carries it to ribosomes for protein synthesis

. 21). What is thioalcohol?

Ans:Thioalcohols (thiols): the oxygen atom of an alcohol is replaced with a sulfur atom

. 22)Why sterilisation is essential in fermentation?

Sterilizing the feed solution is essential because the media cannot contain foreign microbes because this could severely hinder the growth of the production microbe – Most popular method is heat sterilization of the feed solution

23).In which conditions renaturation of DNA takes place?

Ans:Under appropriate conditions (temp. & salt concentration) , separated strands of DNA will renature or reassociate and form the double helix by the process called renaturation (or reannealing)

24)Define gluconeogensis?

Ans :The process of synthesis of glucose from non-carbohydrate sources is known as Gluconeogenesis new formation of sugar, OR Gluconeogenesis is the pathway, which converts pyruvate and related three- and four-carbon compounds to glucose.

25)Write 6 types of fermenter...?

Ans :Simple fermenters (batch and continuous) • Fed batch fermenter • Air-lift or bubble fermenter • Cyclone column fermenter • Tower fermenter • Fluidized bed bioreactors • Packed bed bioreactor

• photobioreactor

26)Glycolysis from pyruvate is via 3 other courts,name them and write on oxidation state..?

Ans:The oxidation state, sometimes referred to as oxidation number, describes degree of oxidation (loss of electrons) of an atom in a chemical compound. The term oxidation was first used by Antoine Lavoisier to signify reaction of a substance with oxygen. Much later, it was realized that the substance, upon

being oxidized, loses electrons, and the meaning was extended to include other reactions in which electrons are lost, regardless of whether oxygen was involved.

27)Define gibbs free energy ?

Ans;Gibbs free energy, G expresses the amount of energy capable of doing work during a reaction at constant temperature and pressure. When a reaction proceeds with the release of free energy (that is, when the system changes so as to possess less free energy), the freeenergy change, G, has a negative value and the reaction is said to be exergonic. In endergonic reactions, the system gains free energy and G is positive. An

28Name of the enzyme require for third bypass of gluconeogenesis?

Ans:Glucose 6 phosphate

29)gluconeogenesis ?

Ans:Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to these other tissues, including brain and muscle, through the bloodstream.

30)Noteworthy chemical transformation during glycolysis?

Ans:(1) degradation of the carbon skeleton of glucose to yield pyruvate; (2) phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, formed during glycolysis (3) transfer of a hydride ion to NAD1, forming NADH

31)Components of fermenter ?

Ans:Formulation of media to be used in culturing the organism during development of inoculum and in the production fermenter 2. Sterilization of the medium, fermenter and ancillary equipment 3. Production of an active, pure culture in sufficient quantity to inoculate the production vessel 4. The growth of the organism in the production fermenter under optimum conditions for product formation 5. The extraction of the product and its purification 6. Disposal of effluents produced by the process.

1. In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP "carboxykinase". 2. The conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from "mitochondria" to the cytosol in the form of malate) 3. When a reaction proceeds with the release of free energy (that is,when the system changes so as to possess less free energy), the freeenergy change, G, has a "negative value" 4. A phosphoryl group (-PO32-) can be transferred to an alcohol (forming a "phosphateester"), or to a carboxylic acid (mixed anhydride) 5. The German biochemist Otto Warburg first observed in 1928 thattumors of nearly all types carry out glycolysis at a much higher rate than normal tissue, even when oxygen is available.This is known as “Warburg effect” 6. A Few Types of "Coenzymes and Proteins" Serve as Universal Electron Carriers 7. about Carbon, Fermentation medium 8. Commensals (Commerial Yeast) Doubt 9. The pathway of glycolysis was explained by Otto Warburg and Hans von Euler-Chelpin in "yeast" 10. ATP consumed and converts in to "ADP+pi" 11. Alcohal is manufactured by "Fermentation" . 12. Hybradization 13:3-D arm: It contains the base "dihydrouridine".

32). What are Thioalcohals?

Ans:Thioalcohols (thiols): the oxygen atom of an alcohol is replaced with a sulfur atom

33) How ribonucleotides held together?

Ans:The ribonucleotides are held together by 3´- 5´ phosphodiester bonds

What is the net yeild of ATP per molecule of glucose and How?

Ans:The net yield is two molecules of ATP per molecule of glucose used, because two molecules of ATP were invested in the prepartory phase. 35

35: Write equation of gluconeogenesis?

Ans;Glucose 6-phosphatase + H20--- Glucose + Pi 36:).

Q36: 5 steps of glycolysis without writing the enzymes name?

Ans: Glycolysis is the metabolic process that serves as the foundation for both aerobic and anaerobic cellular respiration. In glycolysis, glucose is converted into pyruvate. Glucose is a six- memebered ring molecule found in the blood and is usually a result of the breakdown of carbohydrates into sugars. It enters cells through specific transporter proteins that move it from outside the cell into the cell's cytosol. All of the glycolytic enzymes are found in the cytosol. The overall reaction of glycolysis which occurs in the cytoplasm is represented simply as: $C_6H_{12}O_6 + 2\ NAD^+ + 2\ ADP + 2\ P \longrightarrow 2$ pyruvic acid, $(CH_3(C=O)COOH + 2\ ATP + 2\ NADH + 2\ H^+$. 1)The first step in glycolysis is the conversion of D-glucose into glucose-6-phosphate. The enzyme that catalyzes this reaction is hexokinase. 2)The second reaction of glycolysis is the rearrangement of glucose 6-phosphate (G6P) into fructose 6-phosphate (F6P) by glucose phosphate isomerase (Phosphoglucose Isomerase). 3)Phosphofructokinase, with magnesium as a cofactor, changes fructose 6-phosphate into fructose 1,6-bisphosphate. 4)The enzyme Aldolase splits fructose 1, 6-bisphosphate into two sugars that are isomers of each other. These two sugars are dihydroxyacetone phosphate (DHAP) and glyceraldehyde 3-phosphate (GAP). 5)The enzyme triophosphate isomerase rapidly inter- converts the molecules dihydroxyacetone phosphate (DHAP) and glyceraldehyde

3-phosphate (GAP). Glyceraldehyde phosphate is removed / used in next step of Glycolysis.

37) Write steps of synthesis of Phosphoenolpyruvate from Pyruvate?

Ans:The first of the bypass reactions in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP). • This reaction cannot occur by simple reversal of the pyruvate kinase reaction of glycolysis which has a large, negative free-energy change and is irreversible (Table 14–2, step 10) • Instead, the phosphorylation of pyruvate is achieved by a roundabout sequence of reactions that in eukaryotes requires enzymes in both the cytosol and mitochondria. Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine within mitochondria by transamination, in which the alpha -amino group is transferred from alanine (leaving pyruvate) to an alpha-keto carboxylic acid • In mitochondria, pyruvate is converted to oxaloacetate in a biotin (coenzyme)-requiring reaction catalyzed by pyruvate carboxylase, a mitochondrial enzyme

38). What are the components of Eukaryotic Ribosomes. Write about 60s subunit.

The mammalian ribosome contains two major nucleoprotein subunits: • a larger one with 60S

• a smaller one with 40S. The 60S subunit contains • a 5S rRNA • a 5.8S rRNA • a 28S rRNA • more than 50 specific polypeptides. The 40S subunit is smaller and contains • a single 18S rRNA • Approx. 30 distinct polypeptide chains

Q39)Give names of some industrial products fermentation and the products they form in industry. ?

Ans:Yogurt, is produced when the bacterium Lactobacillus bulgaricus ferments the carbohydrate in milk, producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt. Cheese is produced by using Propionibacterium freudenreichii , which ferments milk to produce propionic acid and CO2

; the propionic acid precipitates milk proteins, and bubbles of CO2 cause the holes

Q40). Equation of gluconeogenesis.

Ans :2 pyruvate + 4 ATP + 2 GTP + 2 NADH ====> Glucose + 4 ADP + 2 GDP + 2 NAD+ + 6 inorganic phosphates (Pi)

Q4. Name 2 phases of pentose phosphate pathway.?

There are two phases of the pentose phosphate pathway: 1. The oxidative phase 2. The non-oxidative phase

Q41). Standard reduction porential.?

When two conjugate redox pairs are together in solution, electron

transfer from the electron donor of one pair to the electron acceptor of the other may proceed spontaneously. • The tendency for such a reaction depends on the relative affinity of the electron acceptor of each redox pair for electrons., Measure of this affinity (in volts) is called the standard reduction potential, E

O. **Q42). Which form of pyruvate can immediately tautomerize? What does it form?**

Ans:In this substrate-level phosphorylation, the product pyruvate first appears in its enol form, then tautomerizes rapidly and nonenzymatically to its keto form which predominates at pH 7. form, which predominates at pH 7.

1.Two strands of DNA are_____to each other.(parallel)

2.80% of RNA is_____(transfer RNA)

3.Transformation of RNA into DNA is also known as_____(Reverse transcriptas